



Thapar Institute of Engineering & Technology, Patiala (Deemed to be University)

Department of Electronics & Communication Engineering MST Written Test

BE, ENC

Date: Sept. 27, 2024 (11:00 AM)

Time: 02 Hours; Max. MARKS: 30

UNC302: Object Oriented Programming
Practices

Name of Faculty: Dr. SM, Dr. CMD, Dr. ANS

INSTRUCTIONS:

- Attempt all questions
- Assume any missing information.
- Attempt all the parts of a particular question continuously (for strict compliance)

Questions				Mar ks	CO	BL
Q 1	(a) Design a C++ program for a simple Library Management System. The system should be able to handle the following functionalities, use appropriate classes and objects accordingly: i. Book Management: Add, remove, and search for books. ii. Library Member Management: Add, remove, and search for library members. iii. Borrowing System: Allow members to borrow and return books. Ensure that a book cannot be borrowed if it is already borrowed by another member. (b) Mention any 5 unique difference between Object Oriented Programming and Procedure Oriented Programming.			7	CO 1	L6

13	cout << "Friend Function - Value: " << d.A << endl; }	26	return 0; }
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Answer the following

1. What will be the output of the program?
2. If you uncomment the line number 17 in the NonFriend class, what will happen?
3. Explain the implications of friend functions in C++, particularly regarding encapsulation and access control. How does method overloading impact access specifiers in this example?

Q Given below the following program:

3

Table 2

CO **L3**
2 ,
L4
 ,
L5

<pre> class TMP { int x; public: TMP() { x=0;} TMP(int a) { x=a; } void display() { cout<<"x"<<endl; } // implement and overload pre-decrement operator as a member function // implement and overload post-decrement operator as a member function // implement and overload pre-increment operator as a friend function // implement and overload post-increment operator as a friend function }; int main() { TMP s1(5); s1.display(); --s1; s1.display(); s1--; s1.display(); </pre>	<pre> TMP s2; s2.display(); s2 = --s1; s1.display(); s2.display(); s2 = s1--; s1.display(); s2.display(); s2 = ++s1; s1.display(); s2.display(); s2 = s1++; s1.display(); s2.display(); return 0; } </pre>
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Based on the C++ code mentioned in Table 2, answer the following questions:

- a. Why operator overloading is useful?
- b. In Table 2, observe the C++ code including the MAIN function, and implement the given 4 operator overloading function description using BOLD comments (left side of the Table 2).
- c. Based on the implemented operator overloading functions in 3(a), evaluate the C++ code present in Table 2 and write the output.

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